Akoizo Original Practice - Form 4

Division B | Write It Do It | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (4 points) What is the x coordinate of the blue cube center?

Authored Write It Do It case 4. This is a written reconstruction exercise; coordinates describe the target and are not a diagram handed to a builder in competition. A flat 48-cm square base has an origin at its southwest corner. East is +x and north is +y; coordinates are in cm and viewed from above. Put red cube R at (5,9); blue cube B at (11,9); green cube G at (11,21); and yellow cube Y at (5,21). All cubes have edge 1 cm and centers at the named coordinates. A horizontal straw joins the top centers of R and B, a second joins B and G, and a third joins G and Y. No straw joins Y and R. No parts are hidden. The inventory has four cubes, three straws, and the base.

Answer: _____
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2. (4 points) What is the y coordinate of the green cube center?

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Answer: _____
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3. (4 points) How far east of the red cube center is the blue cube center?

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Answer: _____
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4. (4 points) How far north of the blue cube center is the green cube center?

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Answer: _____
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5. (4 points) What is the straight-line center-to-center distance from red to yellow?

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Answer: _____
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6. (4 points) What is the total center-to-center length of the three specified straw spans?

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7. (4 points) How many straw connections meet the blue cube?

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Answer: _____
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8. (4 points) How many straw connections meet the yellow cube?

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Answer: _____
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9. (4 points) Write unambiguous words-only instructions for establishing the base orientation and placing the red cube.

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10. (4 points) Write a complete words-only sequence to place all four cubes and connect the three straws.

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11. (4 points) A builder places the green cube south of the blue cube. Identify the direction error and give the correction.

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12. (4 points) Explain why “put the other cube beside it” is insufficient for this target.

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13. (4 points) State two checks that catch an unintended mirror image.

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14. (4 points) Describe how you would identify a missing straw without using a drawing in your instructions.

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15. (4 points) Explain the difference between specifying cube centers and specifying the nearest edges.

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16. (4 points) A builder is missing one cube but connects two later pieces correctly to one another. Explain how to document the useful partial reconstruction.

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17. (4 points) Rewrite “left” and “right” in a way that prevents the writer and builder from using opposite viewpoints.

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18. (4 points) Explain how to test the instructions with a partner while avoiding feedback that makes a weak description appear successful.

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